HW #3

1) let $x_0 = 0m$, $V_0 = 0 m/s$, then:

b) $V = V_0 + gt$

$= 0 m/s + 9.81 m/s^2 \cdot 1.55 s$

$= 0 m/s + 15.2055 m/s$

$= 15.2055 m/s \approx 15.21 m/s$

2) $V^2 = V_0^2 + 2g(x - x_0)$

$V^2 - V_0^2 = 2g(x - x_0)$

$\frac{V^2 - V_0^2}{2g} = x - x_0$

$x = \frac{V^2 - V_0^2}{2g} + x_0$

$= \frac{15.21 m/s^2}{2 \cdot 9.81 m/s^2} + 0 m$

$= \frac{231.3441 m^2/s^2}{19.62 m/s^2}$

$\approx 11.79 m$

2) a) $V^2 = V_0^2 + 2g(x - x_0)$

$V = \sqrt{V_0^2 + 2g(x - x_0)}$

$= \sqrt{30.0 m/s^2 + 2 \cdot -9.81 m/s^2 (25m - 0m)}$

$= \sqrt{900 m^2/s^2 - 490.5 m^2/s^2}$

$\approx 20.24 m/s$

b) $\cancel{V = V_0 + gt}$

$\cancel{V - V_0 = gt}$

$\cancel{t = \frac{V - V_0}{g}}$

$\cancel{= \frac{15.0 m/s - 30.0 m/s}{-9.81 m/s^2}}$

$\cancel{= \frac{15.0 m/s}{9.81 m/s^2}}$

$\cancel{\approx 1.53 s}$

b) $+15.0 m/s$

$-15.0 m/s$

c) $V = \sqrt{V_0^2 + 2g(x - x_0)}$

$= \sqrt{900 - 784.8}$

$\approx 10.73 m/s$

3) $x = x_0 + V_0 t + \frac{1}{2} g t^2$

$V_0 t = x - x_0 - \frac{1}{2} g t^2$

$V_0 = \frac{x - x_0 - \frac{1}{2} g t^2}{t}$

$= \frac{0m - 40m + \frac{1}{2} \cdot 9.81 m/s^2 \cdot 4.10 s^2}{4.10 s}$

$= \frac{42.45305 \ m}{4.10 \ s}$

$\approx 10.35 \ m/s$

2) $V = V_0 + at$ // the baseball is at maximum height at the split moment when it changes direction. At the moment, $V = 0 \ m/s$

$at = V - V_0$

$t = \frac{V - V_0}{a}$

$\approx \frac{0 m/s - 10.35 m/s}{-9.81 m/s^2}$

$\approx 1.06 s$

$x = x_0 + V_0 t + \frac{1}{2} g t^2$ // With the given time which the ball needs to reach the top, we can find that height

$= 40m + 10.35 m/s \cdot 1.06 s + \frac{1}{2} \cdot -9.81 m/s^2 \cdot 1.06 s^2$

$= 40m + 10.971 m + 5.511258 m$

$\approx 56.48 \ m$

b) $V^2 = V_0^2 + 2g(x - x_0)$

$V = \sqrt{V_0^2 + 2g(x - x_0)}$

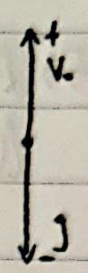
$= \sqrt{10.35 m/s^2 + 2 \cdot -9.81 m/s^2 (25 - 40)}$

$= \sqrt{107.1225 m^2/s^2 + 294.3 m^2/s^2}$

$= \sqrt{401.4225 m^2/s^2}$

$= -20.04 \ m/s$

4) $V = V_0 + at$ // Use the given relation of the velocity at two different instance, time difference, & g to get V_0



$at = V - V_0$

let $z = V_0$

$V = .7z$

$-9.81\,m/s^2 \cdot 1.80\,s = .7z - z$

$-17.658\,m/s = -.3z$

$17.658\,m/s \cdot \frac{10}{3} = z$

$V_0 = z = 58.86\,m/s$

$V = .7z = 41.202\,m/s$

$V = V_0 + at$ // with the initial velocity the following is easy. Given that the projectile reach max height and changes velocity direction, at that split moment, the velocity $= 0\,m/s$

$t = \frac{V - V_0}{a}$

$= \frac{0\,m/s - 58.86\,m/s}{-9.81\,m/s^2}$

$= 6\,s$

$x = x_0 + V_0 t + \frac{1}{2} a t^2$

$= 0m + 58.86\,m/s \cdot 6s + \frac{1}{2} \cdot -9.81\,m/s^2 \cdot 6s^2$

$= 0m + 353.16m + -176.58m$

$= \cancel{178}\,176.58m$ $\quad t = 2 \cdot 6s = 12\,s$

5) ~~$dx_0 + dV_0 t + \frac{1}{2} g t^2 = x$~~

~~$ux_0 + uV_0 t + \frac{1}{2} g t^2 = x$~~

~~$dx_0 + dV_0 t = ux_0 + uV_0 t$~~

~~$= 0m + 25.0\,m/s \cdot t$~~

5) $dV = dV_0 + g\,dt$

$dV_0 = g\,dt + dV$

$= -9.81\,m/s^2 \cdot 1s + 15\,m/s$

$= 5.19\,m/s$

// with the given time, velocity (down) after the given time, we can get the dV_0

2) $dx_0 + dV_0 t + \frac{1}{2} g t^2 = x$

$ux_0 + uV_0 t + \frac{1}{2} g t^2 = x$

$dx_0 + dV_0 t = ux_0 + uV_0 t$

$dV_0 t - uV_0 t = ux_0 - dx_0$

$\cancel{t(dV_0 - uV_0)} = \frac{ux_0 - dx_0}{dV_0 - uV_0}$

$= \frac{0m - 95m}{-5.19m/s - 25m/s}$

$= \frac{95m}{\cancel{14.81m/s}\ 30.19m/s}$

$= \cancel{4.796s}\ 3.147s$

// the height ~~[illegible]~~ and time at two balls meet is the same, so no need for specification. Use algebra to find out we can cancel out both term "$\frac{1}{2} g t^2$" & "x". With the knowledge of the two equation are equal to each other, we can figure out the time takes for two balls to meet.

$x = dx_0 + dV_0 t + \frac{1}{2} g t^2$

$= 95m + 5.19m/s \cdot 3.147s + \frac{1}{2} \cdot -9.81m/s^2 \cdot 3.147s^2$

$= 95m - 16.33293m - 48.57720145m$

$\approx 30.089\ m$

// use either "down" or "up" ball to calculate the meeting position since they share that spot

b) $dV = dV_0 + at$

$= -5.19m/s + -9.81m/s^2 \cdot 3.147s$

$= -5.19m/s - 30.87207\,m/s$

$= \cancel{25.68207\,m/s}$

$36.96207\,m/s$

$uV = uV_0 + at$

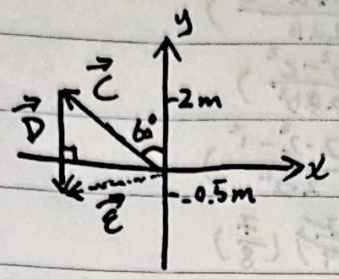
$= 25m/s + -9.81m/s^2 \cdot 3.147s$

$= 25m/s - 30.87207\,m/s$

$= -5.87207\,m/s$

6)

a)



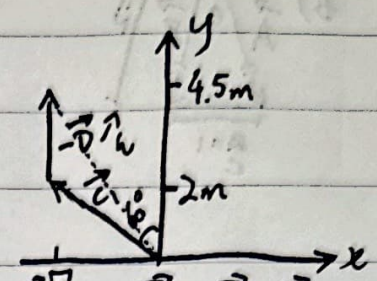
$\vec{E} = \vec{C} + \vec{D}$

$= (C_x, C_y) + (D_x, D_y)$

$= (-2\sqrt{3}, +2) + (0, -2.5)$

$= (-2\sqrt{3}, -.5)$

b)



$\vec{E} = \vec{C} - \vec{D}$

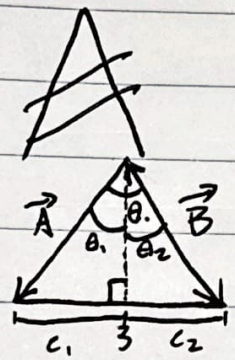
$= (C_x, C_y) - (D_x, D_y)$

$= (-2\sqrt{3}, +2) - (0, -2.5)$

$= (-2\sqrt{3}, 4.5)$

7)

a)



$\because A = 2m$

$B = 2m$

$\therefore A = B$

$\therefore \theta_1 = \theta_2$

$\because \theta_1 = \theta_2$

$A = B$

$\therefore c_1 = c_2$

$\therefore c_1 = c_2 = 1.5\ m$

$\theta_1 = \theta_2 = \arcsin\left(\frac{1.5 \cdot \sin 90°}{2}\right)$

$= \arcsin\left(\frac{3}{4}\right)$

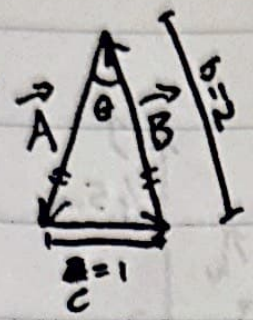
$\approx .848\ RAD \approx 48.6°$

$\theta = \theta_1 + \theta_2$

$= 48.6° \times 2$

$= 97.2°$

b)



$$\Delta\theta = \arccos\left(\frac{b^2+a^2-c^2}{2ab}\right)$$

$$= \arccos\left(\frac{2b^2-c^2}{2b^2}\right)$$

$$= \arccos\left(\frac{2\cdot 2^2-1^2}{2\cdot 2^2}\right)$$

$$= \arccos\left(\frac{7}{8}\right)$$

$$\approx .505 \text{ RAD} \approx 28.29.0^\circ$$

8)

a)

P is at (x_1, y_1)

$r = x\hat{i} + y\hat{j}$

$\hat{i}' = \cos(\theta)\hat{i} + \sin(\theta)\hat{j}$

$\hat{j}' = -\sin(\theta)\hat{i} + \cos(\theta)\hat{j}$

$x' = r\hat{i}'$, $y' = r\hat{j}'$